

Math 520, F07
Quiz 7

Name: Answer
SSN: xxx-xx-

Instructions: There is totally 1 problem. You must show all of your work to receive a credit for a correct answer. You must simplify your answer if possible.

Problem 1. (20 points) Find a particular solution of the equation

$$y'' - 4y' + 4y = (t+1)e^{\alpha t}$$

for the following cases:

(a) $\alpha = -3$.

Solution: characteristic equation is $r^2 - 4r + 4 = 0$
 $\Rightarrow (r-2)^2 = 0$
 $r = 2$.

Thus, the particular solution $x(t)$ has a form of

$$x(t) = e^{-3t}(A_0 + A_1 t)$$

$$x'(t) = e^{-3t}(-3A_0 - 3A_1 t + A_1) = e^{-3t}(-3A_0 + A_1 - 3A_1 t)$$

$$x''(t) = e^{-3t}(9A_0 - 3A_1 + 9A_1 t - 3A_1) = e^{-3t}(9A_0 - 6A_1 + 9A_1 t)$$

$$\Rightarrow L[x](t) = x''(t) - 4x'(t) + 4x(t)$$

$$= e^{-3t}[(9A_0 - 6A_1 + 9A_1 t) - 4(-3A_0 + A_1 - 3A_1 t) + 4(A_0 + A_1 t)]$$

$$= e^{-3t}[25A_0 - 10A_1 + 25A_1 t]$$

$$\Rightarrow \begin{cases} 25A_1 = 1 \\ 25A_0 - 10A_1 = 1 \end{cases} \Rightarrow \begin{cases} A_1 = \frac{1}{25} \\ A_0 = \frac{7}{125} \end{cases}$$

Hence,
$$x(t) = e^{-3t}\left(\frac{7}{125} + \frac{t}{25}\right)$$
$$= \frac{e^{-3t}}{125}(7 + 5t)$$

(b) $\alpha = 2$.

Solution: $\alpha=2$ is the double solution of $r^2-4r+4=0$. Thus

$$x(t) = e^{2t} t^2 (A_0 + A_1 t)$$

$$= e^{2t} V(t)$$

where $V(t) = t^2 (A_0 + A_1 t)$

$$x'(t) = e^{2t} (2V + V')$$

$$x''(t) = e^{2t} (4V + 2V' + 2V' + V'') = e^{2t} (4V + 4V' + V'')$$

$$L[x](t) = x''(t) - 4x'(t) + 4x(t)$$

$$= e^{2t} V''(t)$$

$$\Rightarrow V''(t) = t + 1$$

$$V'(t) = \int t + 1 dt = \frac{t^2}{2} + t$$

$$V(t) = \int V'(t) dt = \int \frac{t^2}{2} + t dt = \frac{t^3}{6} + \frac{t^2}{2}$$

Hence
$$x(t) = e^{2t} \left(\frac{t^3}{6} + \frac{t^2}{2} \right)$$
$$= \frac{e^{2t}}{6} (t^3 + 3t^2)$$